
Correlation

Lecture 5: Fatemeh Riahi

Correlation Analysis

Two factors are correlated when values of x has some predictive power on the value of y .

The **correlation coefficient** of X and Y measures the degree to which Y is a function of X (and visa versa).

Correlation ranges from -1 (anti-correlated) to 1 (fully correlated) through 0 (uncorrelated).

The Pearson Correlation Coefficient

The numerator defines the **covariance**, which determines the sign but not the scale.

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

A point (x,y) makes a positive contribution to r when both are above or below their means.

Representative Pearson Correlations

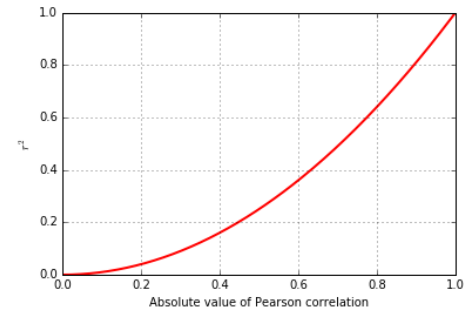
- SAT scores and freshman GPA ($r=0.47$)
 - SAT scores and economic status ($r=0.42$)
 - Income and coronary disease ($r=-0.717$)
 - Smoking and mortality rate ($r=0.716$)
 - Video games and violent behavior ($r=0.19$)
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Interpreting Correlations: r^2

The square of the sample correlation coefficient r^2 estimates the fraction of the variance in Y explained by X in a simple linear regression.

Thus the predictive value of a correlation decreases quadratically with r .

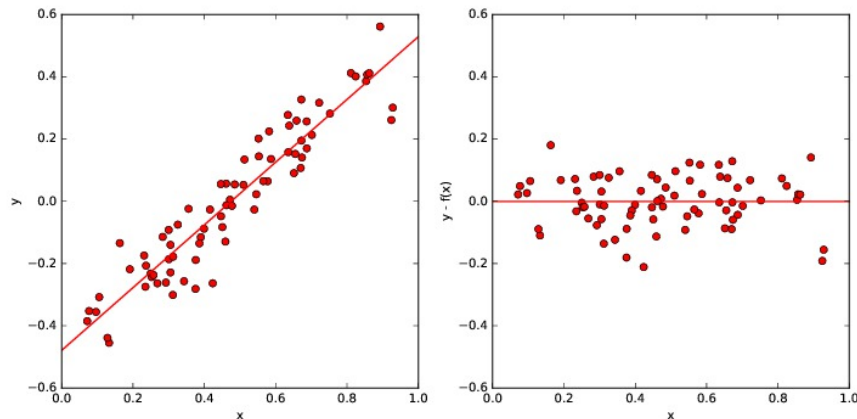
The correlation between height and weight is approximately 0.8, meaning it explains about $\frac{2}{3}$ of the variance.



Variance Reduction and “ R^2 ”

If there is a good linear fit $f(x)$, then the residuals $y - f(x)$ will have lower variance than y .

Generally speaking,
 $1 - r^2 = V(r) / V(y)$
Here $r = 0.94$,
explaining 88.4% of
 $V(y)$.

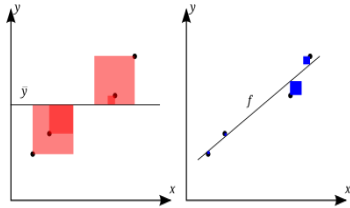


Coefficient of Determination

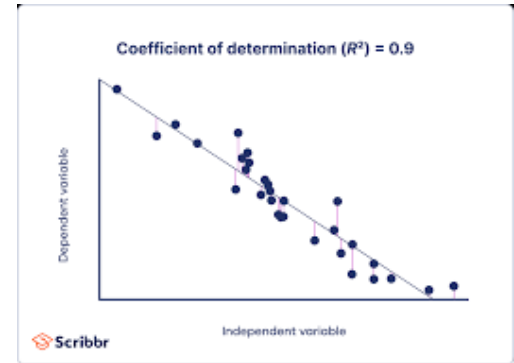
SS_{res} = residual sum of squares error $(y - f(x))^2$

SS_{tot} = total sum of squares

R^2 = coefficient of determination



$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$



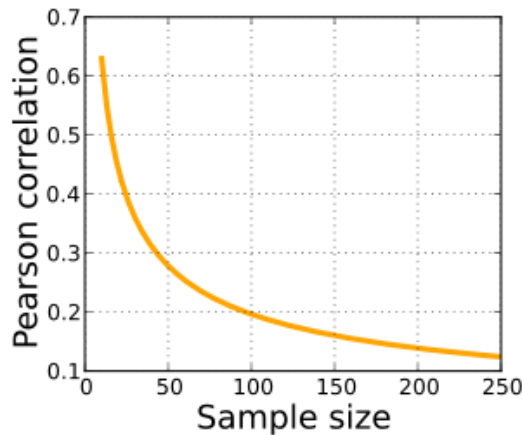
Models better than **linear** regression can produce a better R^2 .

Interpreting Correlation: Significance

The statistical significance of a correlation depends upon the sample size as well as r .

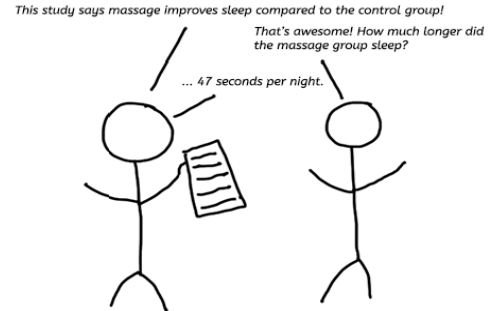
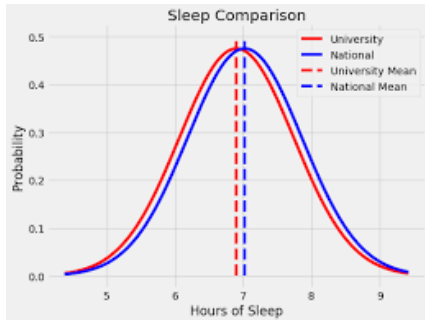
Even small correlations become significant (at the 0.05 level) with large-enough sample sizes.

This motivates “big data” multiple parameter models: each single correlation may explain/predict only small effects, but large numbers of weak but *independent* correlations may together have strong predictive power.



The Significance of Significance

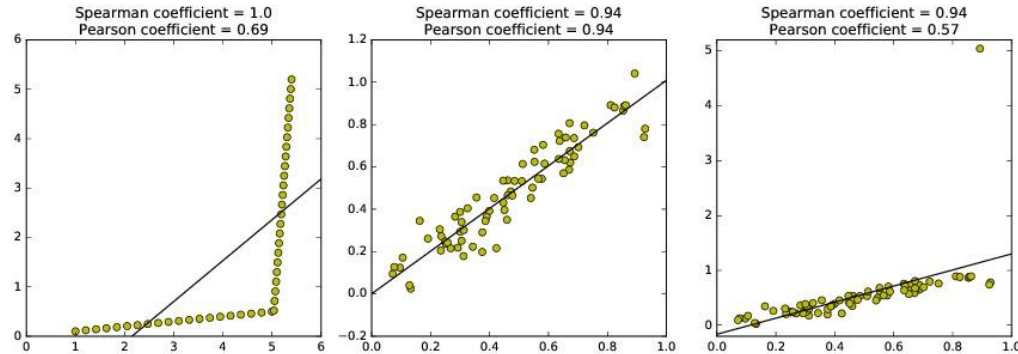
Significance testing measures if there is a “real” relationship, instead of being random chance. It does not measure whether the relationship is important, typically called “effect size”.



Spearman Rank Correlation

Counts the number of disordered pairs, not how well the data fits a line.

Thus better with non-linear relationships and outliers.



Computing Spearman Correlation

Let $rank(x_i)$ be the rank position of x_i in sorted order, from 1 to n . Then:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

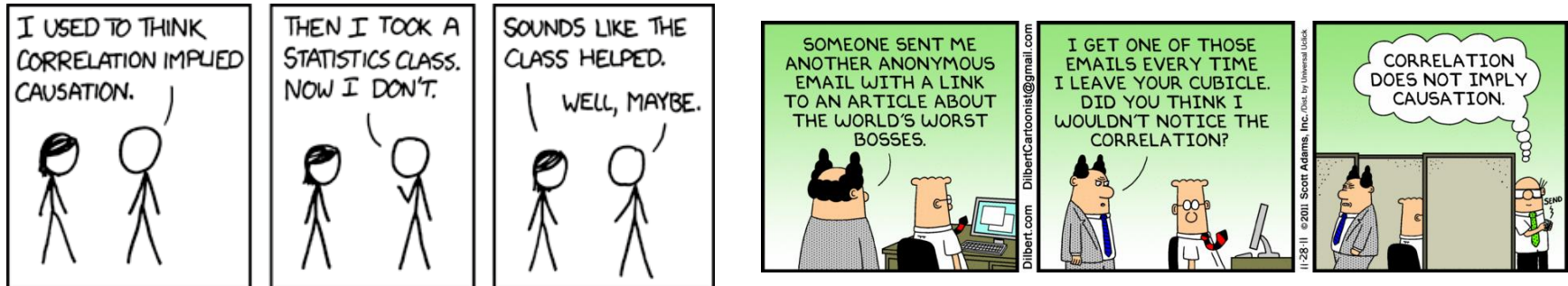
where $d_i = rank(x_i) - rank(y_i)$.

It is the Pearson correlation of the X and Y value ranks, so it ranges from -1 to 1.

Correlation vs. Causation

Correlation does not mean causation.

The number of police active in a precinct correlated strongly with the local crime rate, but the police do not cause the crime.

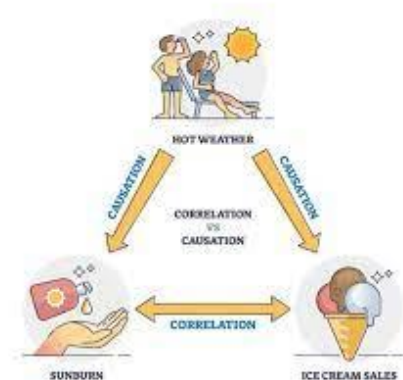
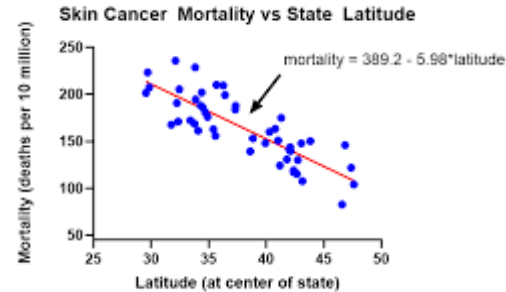


Possible A/B Correlation Explanations

- A causes B
- B causes A
- C causes both A and B
- Spurious correlations

(butter production in Bangladesh)

Social scientists are very good at
not ascribing causality to correlations



Autocorrelation and Periodicity

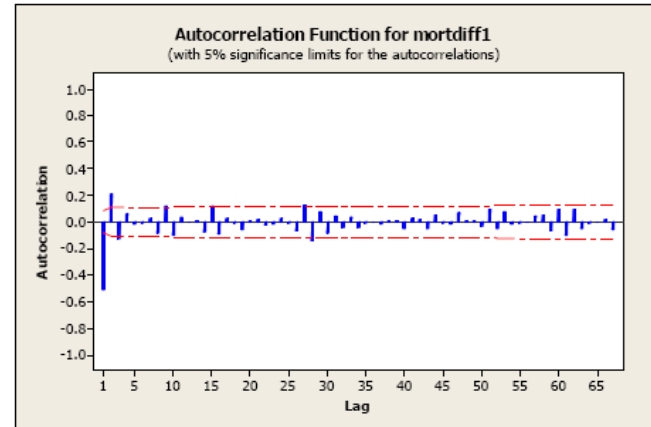
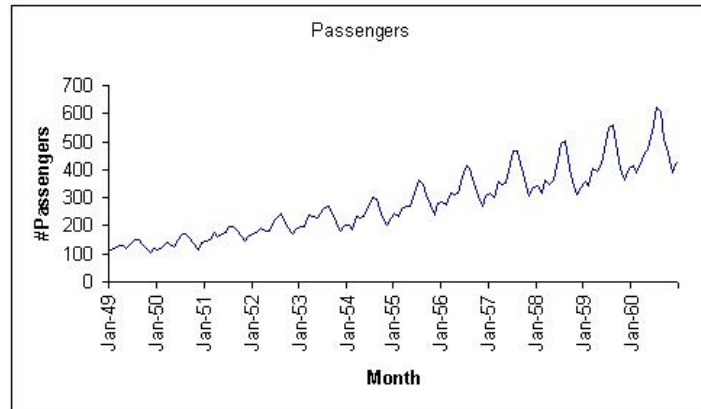
Time-series data often exhibits cycles which affect its interpretation.

Sales in different businesses may well have 7 day, 30 day, 365 day, and 4×365 day cycles.

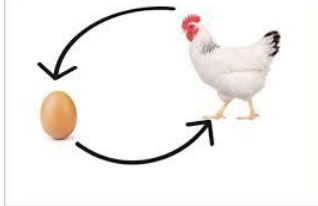
A cycle of length k can be identified by unexpectedly large autocorrelation between $S[t]$ and $S[t+k]$ for all $0 < t < n-k$.

The Autocorrelation Function

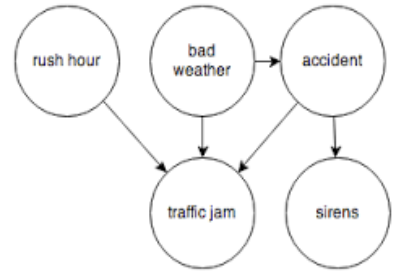
Computing the lag-k autocorrelation takes $O(n)$, but the full set can be computed in $O(n \log n)$ via the Fast Fourier Transform (FFT).



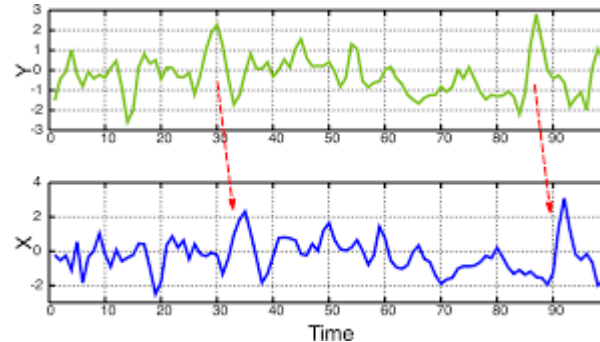
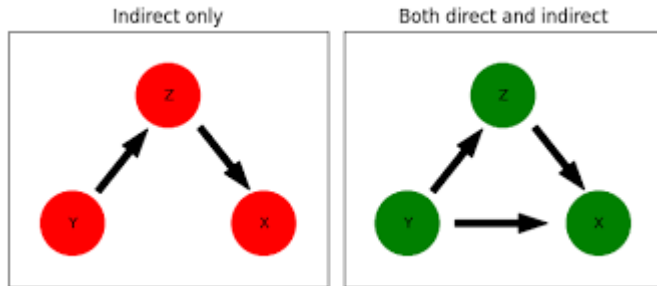
Granger Causality Testing



Y “Granger causes” X if there is greater correlation with a time lag.



Says nothing about direct vs. indirect causes.



Did his causality test win Granger the Nobel Prize?

Logarithms

The logarithm is the inverse exponential function, i.e. $y = \log_b x \implies b^y = x$

We will use them here for reasons different than in algorithms courses:

Summing logs of probabilities is more numerically stable than multiplying them:

$$\prod_{i=1}^n p_i = b^P \text{ where } P = \sum_{i=1}^n \log_b(p_i)$$

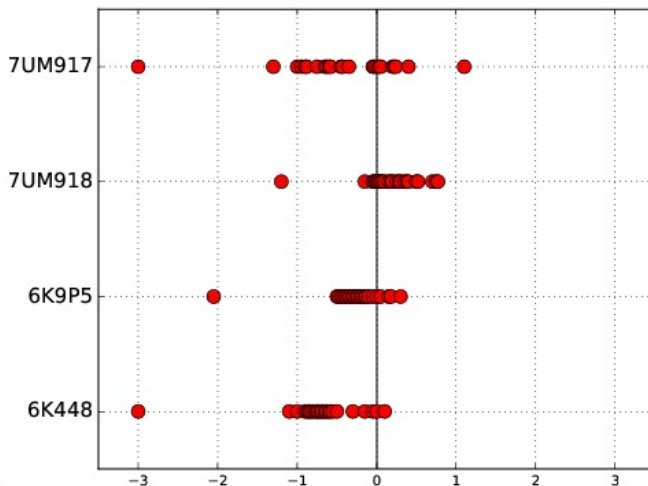
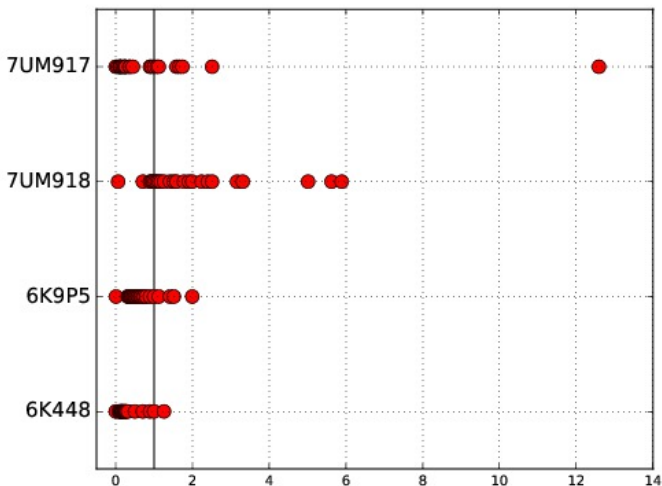
Logarithms and Ratios

Ratios of two similar quantities (e.g. $\text{new_price} / \text{old_price}$) behave differently when reflecting increases vs. decreases.

200/100 is 200% above baseline, but 100/200 is 50% below despite being similar changes!

Taking the log of the ratios yield equal displacement: 1.0 and -1.0 (for base-2 logs)

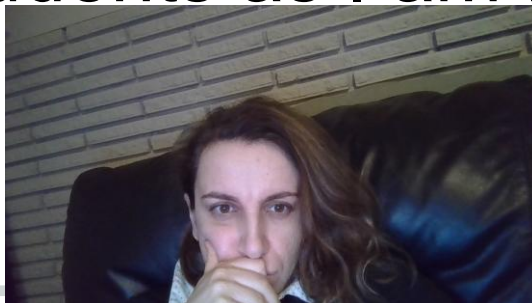
Always Plot Logarithms of Ratios!



Logarithms and Power Laws

Taking the logarithm of variables with a power law distribution brings them more in line with traditional distributions.

My wealth is roughly the same number of logs from typical students as I am from Bill Gates!



Normalizing Skewed Distributions

Taking the logarithm of a value before analysis is useful for power laws and ratios.

